



We can talk in general terms about the **development of a baby**—when organs grow, when features form, when she might be ready for birth—but the only certainty is that **your baby will be unique**. From the wave of her hair to her toe prints, she won't have a single **characteristic** that's exactly the same as any other living being.

My baby is unique

What makes babies unique?

Uniqueness comes from two main sources. Firstly, before your baby is born, the “soup” of chromosomes that she gets from each parent gives her a set of genes that might switch on or off, both in the uterus and after birth. Secondly, your baby's environment (for example, from where you live to your morals and values) molds her view and experience of the world. This leads to her responding to her environment in a conscious way—called learned behavior—but it also has physical effects in her body, mutating her genetic codes so that she can adapt to life. In primitive times, this was essential to survival.

Genes are the building blocks of DNA—the hereditary blueprint that's in every cell in your baby's body. Genes give her her eye and hair color, her fingerprints, her inherited character traits, and her inherited diseases and conditions. They're embedded in the chromosomes she receives from her parents. Current scientific thinking is that your baby (and every human being) has around 24,000 genes attached to 23 pairs of chromosomes. Aside from the chromosomes that determine gender, every chromosome has two copies of each gene. Every gene is also made up of variations of itself. Variations of the same gene are called alleles, and these can be recessive alleles or dominant ones. The dominant alleles “overpower” the

recessive ones, producing a unique physical, mental, and emotional recipe for every human being.

The brain and the environment

Genes provide the raw ingredients, but other aspects of your baby's uniqueness—such as her personality, how she learns to do things, the way she responds to stimuli in her environment, how she displays her intellect, and the ways in which she forms relationships—are subject to her experience of the world around her.

From the moment of her birth, the information that your baby gets from her environment builds connections in her brain. At birth, brain imaging shows large sections of the brain waiting to form neural connections. These come with such speed that by 24 months, only small unconnected patches remain. By the age of three, your baby will have far more connections than she ever needs and a period of “synaptic pruning” begins. This eliminates and hones the connections so that brain efficiency rises.

The role of lifestyle and its influence on our genes is important when considering inherited illness. For example, we know that heart disease can also be inherited. However, while a baby might inherit a tendency toward it, her lifestyle choices can turn on or off the gene that causes the condition.